Exercise J-M

1. Let $y = y(x)$ be the solution of the differential equation

$$x\tan\left(\frac{y}{x}\right)dy = \left(y\tan\left(\frac{y}{x}\right) - x\right)dx,\ -1 \le x \le 1, y\left(\frac{1}{2}\right) = \frac{\pi}{6}$$

.

Then the area of the region bounded by the curves $x = 0, x = \frac{1}{\sqrt{2}}$ and $y = y(x)$ in the upper half plane is :

[JEE(Main)July2021]

माना अवकल समीकरण

$$x\tan\left(\frac{y}{x}\right)dy = \left(y\tan\left(\frac{y}{x}\right) - x\right)dx,\ -1 \le x \le 1, y\left(\frac{1}{2}\right) = \frac{\pi}{6}$$

का हल $y = y(x)$ है। तो वक्रों $x = 0, x = \frac{1}{\sqrt{2}}$ तथा $y = y(x)$ द्वारा ऊपरी आधे निर्देशांक तल में घिरे क्षेत्र का क्षेत्रफल है :

[JEE(Main)July2021]

(A) $\frac{1}{8}(\pi - 1)$ (B) $\frac{1}{12}(\pi - 3)$ (C) $\frac{1}{4}(\pi - 2)$ (D) $\frac{1}{6}(\pi - 1)$

Ans. A

Sol. $\tan\left(\frac{y}{x}\right)\frac{dy}{dx} = \frac{y}{x}\tan\left(\frac{y}{x}\right) - 1$

Put $y = vx$

$$\frac{dy}{dx} = v + \frac{xdv}{dx}$$

$$(\tan v)\left(v + \frac{xdv}{dx}\right) = v\tan v - 1$$

$$v\tan v + \left(\frac{xdv}{dx}\right)\tan v = v\tan v - 1$$

$$\int \tan v\, dv = -\int \frac{dx}{v}$$

$$\ln(\sec v) = -\ln x + \ln c$$

$$\sec v = \frac{c}{x}$$

$$\sec\left(\frac{y}{x}\right) = \frac{c}{x}$$

$x = 1/2 \Rightarrow y = \pi/6$

$\Rightarrow c = 1$

$$\cos\left(\frac{y}{x}\right) = x$$

$$y = x\cos^{-1}(x)$$

$$\text{Area} = \int_0^{1/\sqrt{2}} x\cos^{-1}(x)\,dx$$

$$x = \cos\theta d\theta$$

$$dx = -\sin\theta\, d\theta$$

$$= \frac{1}{4}\int_{\pi/2}^{\pi/2} t\sin t\,dt$$

$$= \frac{1}{8}\left((-t\cos t)_{\pi/2} + \int_{\pi/2}^{\pi}\cos t dt\right)$$

$$= \frac{1}{8}\left((\pi) + (\sin t)_{\pi/2}^{\pi}\right)$$

$$= \frac{(\pi-1)}{8}$$

2. The area (in sq. units) of the region, given by the set $\{(x, y) \in R \times R \mid x \geq 0, 2x^2 \leq y \leq 4 - 2x\}$ is :

[JEE(Main)July2021]

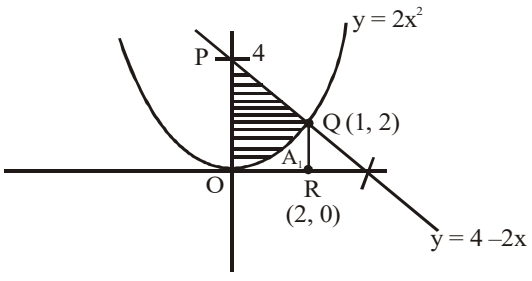
क्षेत्र $\{(x, y) \in R \times R \mid x \geq 0, 2x^2 \leq y \leq 4 - 2x\}$ का क्षेत्रफल (वर्ग इकाइयों में) बराबर है :

[JEE(Main)July2021]

(A) $\frac{13}{3}$ (B) $\frac{17}{3}$ (C) $\frac{7}{3}$ (D) $\frac{8}{3}$

Ans. C

Sol.



$$A_1 = \int_0^1 2x^2 dx = \left(\frac{2}{3}x^3\right)_0^1 = \frac{2}{3}$$

Req Area = Area of trapezium ORQP – A_1

$$= \frac{1}{2}(4+2)(1) - \frac{2}{3}$$

$$= 3 - \frac{2}{3} = \frac{7}{3}$$

3. If the area of the bounded region

$$R = \left\{ (x, y) : \max\{0, \log_e x\} \le y \le 2^x, \frac{1}{2} \le x \le 2 \right\}$$

is $\alpha(\log_e 2)^{-1} + \beta(\log_e 2) + \gamma$, then value of $(\alpha + \beta - 2\gamma)^2$ is equal to : **[JEE(Main)July2021]**

यदि परिबद्धित क्षेत्र

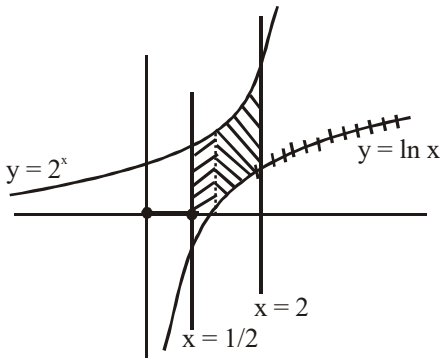
$$R = \left\{ (x, y) : \max\{0, \log_e x\} \le y \le 2^x, \frac{1}{2} \le x \le 2 \right\}$$

का क्षेत्रफल $\alpha(\log_e 2)^{-1} + \beta(\log_e 2) + \gamma$ है, तो $(\alpha + \beta - 2\gamma)^2$ का मान बराबर है : **[JEE(Main)July2021]**

(A) 4 (B) 8 (C) 1 (D) 2

Ans. D

Sol.



$$A = \int_{\frac{1}{2}}^{1} 2^x dx + \int_{1}^{2} \left(2^x - \ln x\right) dx$$

$$A = \frac{2^x}{\ln 2}\Big|_{\frac{1}{2}}^{2} + \frac{2^x}{\ln 2} - \left(x \ln x - x\right)\Big|_{1}^{2}$$

$$A = \left(4 - \sqrt{2}\right)\log_2 e - 2\log_2 e + 1$$

$$\alpha = 4 - \sqrt{2}, \beta = -2, \gamma = 1$$

$$(\alpha + \beta - 2\gamma)^2 = 2$$

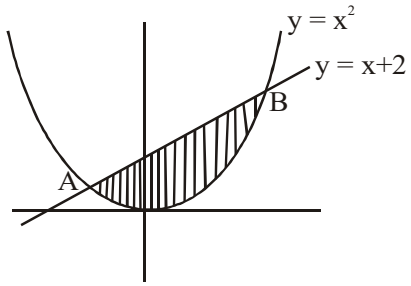
4. The area of the region bounded by $y - x = 2$ and $x^2 = y$ is equal to : **[JEE(Main)July2021]**

$y - x = 2$ तथा $x^2 = y$ द्वारा घिरे क्षेत्र का क्षेत्रफल बराबर है : **[JEE(Main)July2021]**

(A) $\frac{16}{3}$ (B) $\frac{4}{3}$ (C) $\frac{2}{3}$ (D) $\frac{9}{2}$

Ans. D

Sol.



For x co-ordinate of A and B

$x^2 = x + 2$

$\Rightarrow x^2 - x - 2 = 0$

$(x-2)(x+1) = 0$

$\therefore x = -1, 2$

Required Area :

$$= \int_{-1}^{2} \left((x+2) - x^2\right) dx$$

$$= \left[\frac{x^2}{2} + 2x - \frac{x^3}{3}\right]_{-1}^{2}$$

$$= \left[2 + 4 - \frac{8}{3}\right] - \left[\frac{1}{2} - 2 + \frac{1}{3}\right]$$

$$= \frac{10}{3} - \left[\frac{3 - 12 + 2}{6}\right]$$

$$= \frac{10}{3} - \left[\frac{3 - 12 + 2}{6}\right]$$

$$= \frac{9}{2}$$

5. Let T be the tangent to the ellipse E : $x^2 + 4y^2 = 5$ at the point P (1, 1). If the area of the region bounded by the tangent T, ellipse E, lines $x = 1$ and $x = \sqrt{5}$ is $\alpha\sqrt{5} + \beta + \gamma\cos^{-1}\left(\frac{1}{\sqrt{5}}\right)$, then $|\alpha + \beta + \gamma|$ is equal to _____.

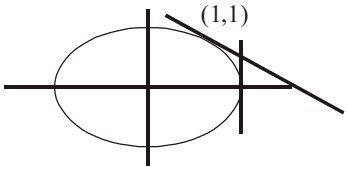
[JEE(Main)July2021]

माना दीर्घवृत E : $x^2 + 4y^2 = 5$ के बिंदु P (1, 1) पर स्पर्श रेखा T है। यदि स्पर्श रेखा T, दीर्घवृत E तथा रेखाओं x = 1 और $x = \sqrt{5}$ द्वारा घिरे क्षेत्र का क्षेत्रफल $\alpha\sqrt{5} + \beta + \gamma\cos^{-1}\left(\frac{1}{\sqrt{5}}\right)$ है, तो $|\alpha + \beta + \gamma|$ बराबर है _____

[JEE(Main)July2021]

Ans. 1.25

Sol.



E : $x^2 + 4y^2 - 5$

T : $x + 4y = 5$

$$\text{Area} = \int_1^{\sqrt{5}} (y_{\text{LINE}} - y_{\text{ELLIPSE}})\,dx$$

$$= \int_1^{\sqrt{5}} \left(\frac{5-x}{4} - \sqrt{\frac{5-x^2}{4}}\right) dx$$

$$= \frac{1}{4}(5x)_1^{\sqrt{5}} - \left(\frac{x^2}{8}\right)_1^{\sqrt{5}} - \frac{1}{4}\left(x\sqrt{5-x^2}\right)_1^{\sqrt{5}} - \frac{5}{4}\left(\sin^{-1}\left(\frac{x}{\sqrt{5}}\right)\right)_1^{\sqrt{5}}$$

$$= \frac{5\sqrt{5}}{4} - \frac{5}{4} - \frac{1}{2} - \frac{1}{4}(0-2) - \frac{5}{4}\left(\frac{\pi}{2} - \sin^{-1}\left(\frac{1}{\sqrt{5}}\right)\right)$$

$$= \frac{5\sqrt{5}}{4} - \frac{5}{4} - \frac{5}{4}\cos^{-1}\left(\frac{1}{\sqrt{5}}\right)$$

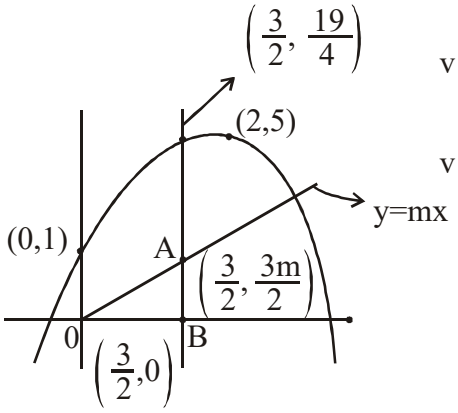
$$|\alpha + \beta + \gamma| = \frac{5}{4}$$

6. If the line y = mx bisects the area enclosed by the lines x = 0, y = 0, $x = \frac{3}{2}$ and the curve $y = 1 + 4x - x^2$, then 12 m is equal to_____. **[JEE(Main)Aug.2021]**

यदि रेखाओं x = 0, y = 0, $x = \frac{3}{2}$ तथा वक्र $y = 1 + 4x - x^2$ से घिरे क्षेत्र के क्षेत्रफल को रेखा y = mx समद्विभाजित करती है, तो 12 m बराबर है _____ **[JEE(Main)Aug.2021]**

Ans. 26

Sol.



$$\text{Total Area} = \int_0^{3/2} (1 + 4x - x^2)dx$$

$$= x + 2x^2 - \frac{x^3}{3}\Big|_0^{3/2}$$

$$= \frac{3}{2} + 2\left(\frac{9}{4}\right) - \frac{1}{3}\left(\frac{27}{8}\right)$$

$$= \frac{3}{2} + \frac{9}{2} - \frac{9}{8}$$

$$= \frac{12 + 36 - 9}{8} = \frac{39}{8}$$

$$\text{Area of } \Delta OAB = \frac{39}{16}$$

$$\frac{39}{16} = \frac{1}{2} \times \frac{3}{2} \times \frac{3m}{2}$$

$$\frac{13}{2} = 3m$$

$$12m = \frac{13}{2} \times 4 = 26$$

7. The area (in sq. units) of the region bounded by the curves $x^2 + 2y - 1 = 0$, $y^2 + 4x - 4 = 0$ and $y^2 - 4x - 4 = 0$, in the upper half plane is _____. **[JEE(Main)July2021]**

ऊपरी आधे निर्देशांक तल में वक्रों, $x^2 + 2y - 1 = 0$, $y^2 + 4x - 4 = 0$ तथा $y^2 - 4x - 4 = 0$ द्वारा परिबद्ध क्षेत्र का क्षेत्रफल (वर्ग इकाइयों में) बराबर है _____ **[JEE(Main)July2021]**

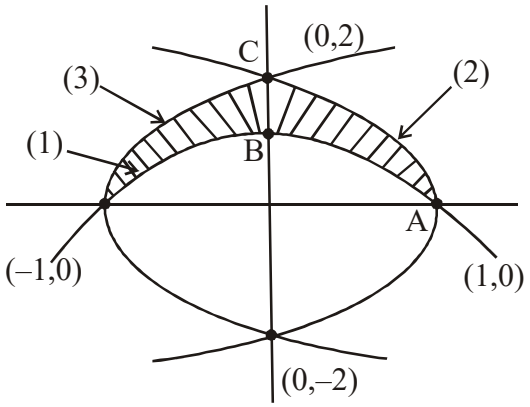
Ans. 2

Sol. $x^2 = -2y + 1$

$x^2 = -2(y - \frac{1}{2})$ .....(1)

$y^2 = -4(x - 1)$ .....(2)

$y^2 = 4(x + 1)$ .....(3)



Required area = 2 (area ABC)

$$= 2\int_0^1 \left(2\sqrt{1-x} - \frac{(1-x^2)}{2} \right) dx$$

= 2 square unit.

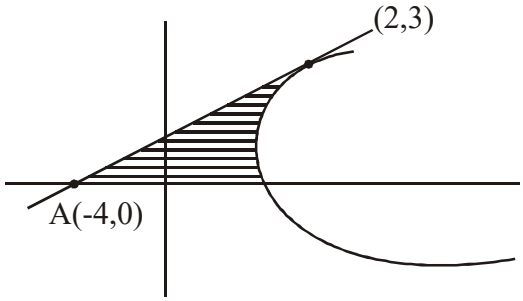
8. The area of the region bounded by the parabola $(y-2)^2 = (x-1)$, the tangent to it at the point whose ordinate is 3 and the x-axis is : **[JEE(Main)Aug.2021]**

परवलय $(y-2)^2 = (x-1)$, इसके उस बिन्दु जिसकी कोटि 3 है पर स्पर्श रेखा तथा x-अक्ष द्वारा परिबद्ध क्षेत्र का क्षेत्रफल है : **[JEE(Main)Aug.2021]**

(A) 6 (B) 4 (C) 9 (D) 10

Ans. C

Sol. $(y-2)^2 = (x-1)$



slope of tangent

$$2(y-2)\frac{dy}{dx} = 1$$

$$\frac{dy}{dx} = \frac{1}{2}$$

tangent $y - 3 = \frac{1}{2}(x-2)$

$2y - 6 = x - 2$

$x = 2y - 4$

Required area $= \int_0^3 \left[(y-2)^2 + 1 - (2y-4) \right] dy$

$$= \int_0^3 (y-2)^2 - 2y + 5 \; dy$$

$$= \frac{(y-2)^3}{3} - y^2 + 5y \Big]_0^3$$

$= \frac{1}{3} - 9 + 15 + \frac{8}{3}$

$= 6 + 3$

$= 9$ square unit

9. The area, enclosed by the curves $y = \sin x + \cos x$ and $y = |\cos x - \sin x|$ and the lines $x = 0,\ x = \frac{\pi}{2}$, is :

[JEE(Main)Aug.2021]

वक्रों $y = \sin x + \cos x$ एवं $y = |\cos x - \sin x|$ तथा रेखाओं $x = 0,\ x = \frac{\pi}{2}$ से घिरे क्षेत्र का क्षेत्रफल है :

[JEE(Main)Aug.2021]

(A) $2\sqrt{2}\left(\sqrt{2}+1\right)$ (B) $2\left(\sqrt{2}+1\right)$ (C) $2\sqrt{2}\left(\sqrt{2}-1\right)$ (D) $4\left(\sqrt{2}-1\right)$

Ans. C

Sol. $y = \sin x + \cos x$

$y = |\cos x - \sin x|$

For $x \in \left(0, \frac{\pi}{2}\right), (\sin x + \cos x) > |\cos x - \sin x|$

Required Area

$$= \int_0^{\frac{\pi}{2}} (\sin x + \cos x - |\cos x - \sin x|)dx$$

$$= \int_0^{\frac{\pi}{2}} (\sin x + \cos x - (\cos x - \sin x|)dx$$

$$+ \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} (\sin x + \cos x - (\sin x - \cos x|)dx$$

$$= \int_0^{\frac{\pi}{4}} (2\sin x)dx + \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} (2\cos x)dx$$

$$= 2\left[\left[-\cos x\right]_0^{\frac{\pi}{4}}\right] + 2\left[\left[\sin x\right]_{\frac{\pi}{4}}^{\frac{\pi}{2}}\right]$$

$$= 2\left[\left[-\frac{1}{\sqrt{2}}+1\right]\right]+2\left[1-\frac{1}{\sqrt{2}}\right]$$

$$= -\sqrt{2}+2+2-\sqrt{2}$$

$$= 4-2\sqrt{2}$$

$$= 2(2-\sqrt{2})$$

$$= 2\sqrt{2}(\sqrt{2}-1)$$

10. The area of the region $S=\{(x,y):3x^2 \le 4y \le 6x+24\}$ is _____. **[JEE(Main)Aug.2021]**

क्षेत्र $S=\{(x,y):3x^2 \le 4y \le 6x+24\}$ का क्षेत्रफल है_____ **[JEE(Main)Aug.2021]**

Ans. 27

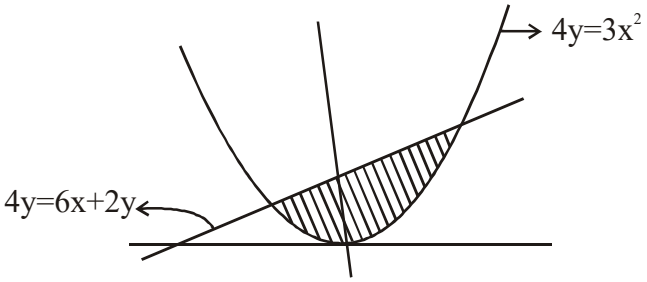
Sol. POM of intersection

$3x^2 = 6x + 24$

$\Rightarrow x = 4, x = -2$

$$\text{Area} = \int_{-2}^{4}\left(\frac{6x+24}{4}-\frac{3x^2}{4}\right)dx$$

$= 27$



11. Let a and b respectively be the points of local maximum and local minimum of the function $f(x) = 2x^3 - 3x^2 - 12x$. If A is the total area of the region bounded by $y = f(x)$, the x-axis and lines $x = a$ and $x = b$, then 4A is equal to _____. **[JEE(Main)Aug.2021]**

माना फलन $f(x) = 2x^3 - 3x^2 - 12x$ के स्थानीय उच्चतम तथा स्थानीय निम्नतम बिन्दु क्रमशः a तथा b है। यदि $y = f(x)$, x-अक्ष, तथा रेखाओं $x = a$ और $x = b$ से घिरे क्षेत्र का क्षेत्रफल A है, तो 4A बराबर है_____ **[JEE(Main)Aug.2021]**

Ans. 114

Sol. $f(x) = 2x^3 - 3x^2 - 12x$

$f'(x) = 6x^2 - 6x - 12$

$= 6(x+1)(x-2)$

$x = -1, 2$

$$A = \int_{-1}^{0} \left(2x^3 - 3x^2 - 12x\right) dx - \int_{0}^{2} \left(2x^3 - 3x^2 - 12x\right) dx$$

$$A = \left|\frac{x^4}{2} - x^3 - 12\right)_{-1}^{0} - \left(\frac{x^4}{2} - x^3 - 12\right)_{0}^{2}$$

$$A = \frac{57}{2} \Rightarrow 4A = 114$$

12. The area of the region enclosed between the parabolas $y^2 = 2x - 1$ and $y^2 = 4x - 3$ is :

[JEE(Main)June2022]

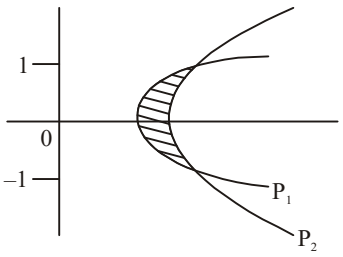
परवलयों $y^2 = 2x - 1$ तथा $y^2 = 4x - 3$ के बीच घिरे क्षेत्र का क्षेत्रफल है : **[JEE(Main)June2022]**

(A) $\frac{1}{3}$ (B) $\frac{1}{6}$ (C) $\frac{2}{3}$ (D) $\frac{3}{4}$

Ans. A

Sol. $P_1 : y^2 = 2x - 1$

$P_2 : y^2 = 4x - 3$



$$A = 2.\int_{0}^{1} \left(x_{P_2} - x_{P_1}\right) dy$$

$$A = 2\int_{0}^{1} \left(\frac{y^2+3}{4}\right) - \left(\frac{y^2+1}{2}\right) dy$$

$$A = \frac{1}{3}$$

13. The area bounded by the curve $y = |x^2 - 9|$ and the line $y = 3$ is : **[JEE(Main)June2022]**

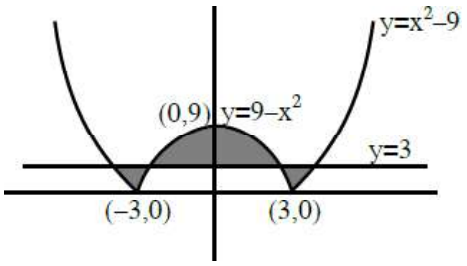
वक्र $y = |x^2 - 9|$ तथा रेखा $y = 3$ द्वारा घिरे क्षेत्र का क्षेत्रफल है : **[JEE(Main)June2022]**

(A) $4\left(2\sqrt{3} + \sqrt{6} - 4\right)$ (B) $4\left(4\sqrt{3} + \sqrt{6} - 4\right)$

(C) $8\left(4\sqrt{3} + 3\sqrt{6} - 9\right)$ (D) $8\left(4\sqrt{3} + 2\sqrt{6} - 9\right)$

Ans. D

Sol.



Area of shaded region

$$= 2\int_0^3 \left(\sqrt{9+y} - \sqrt{9-y}\right)dy + 2\int_3^9 \left(\sqrt{9-y}\right)dy$$

$$= 2\left[\int_0^3 (9+y)^{1/2} - \int_0^3 (9-y)^{1/2}\,dy + \int_3^9 \left(\sqrt{9-y}\right)^{1/2} dy\right]$$

$$= 2\left[\frac{2}{3}\left[(9+y)^{3/2}\right]_0^3 + \frac{2}{3}\left[(9-y)^{3/2}\right]_0^3 - \frac{2}{3}\left[(9-y)^{3/2}\right]_3^9\right]$$

$$= \frac{4}{3}\left[12\sqrt{12} - 27 + 6\sqrt{6} - 27 - \left(0 - 6\sqrt{6}\right)\right]$$

$$= \frac{4}{3}\left[24\sqrt{3} + 12\sqrt{6} - 54\right]$$

$$= 8\left(4\sqrt{3} + 2\sqrt{6} - 9\right)$$

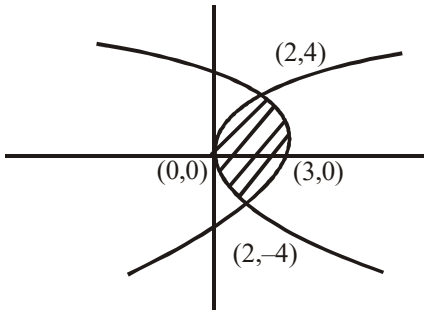
14. The area of the region bounded by $y^2 = 8x$ and $y^2 = 16(3 - x)$ is equal to : **[JEE(Main)June2022]**

वक्रों $y^2 = 8x$ तथा $y^2 = 16(3 - x)$ द्वारा घिरे क्षेत्र का क्षेत्रफल है : **[JEE(Main)June2022]**

(A) $\frac{32}{3}$ (B) $\frac{40}{3}$ (C) 16 (D) 19

Ans. C

Sol.



$$8x = 16(3 - x)$$

$x = 6 - 2x$

$3x = 6 \Rightarrow x = 2$

$y^2 = 16 \Rightarrow y = \pm 4$

$x = y^2/8,$

$x = \dfrac{48 - y^2}{16}$

$$\text{Area} = 2\int_0^4 \left(\left(\frac{48 - y^2}{16} \right) - \left(\frac{y^2}{8} \right) \right) dy = 16$$

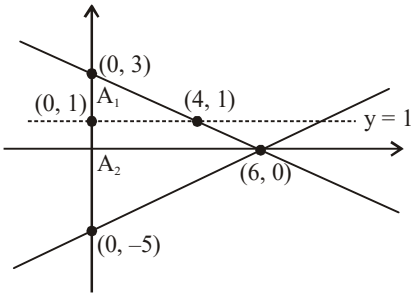
15. If a point A (x,y) lies in the region bounded by the y-axis, straight lines $2y + x = 6$ and $5x - 6y = 30$, then the probability that $y < 1$ is : **[JEE(Main)June2022]**

यदि y-अक्ष, सरल रेखाओं $2y + x = 6$ तथा $5x - 6y = 30$ से घिरे (bounded) क्षेत्र में एक बिन्दु A(x, y) है, तो $y < 1$ होने की प्रायिकता है : **[JEE(Main)June2022]**

(A) $\frac{1}{6}$ (B) $\frac{5}{6}$ (C) $\frac{2}{3}$ (D) $\frac{6}{7}$

Ans. B

Sol.



$$P(y < 1) = \frac{A_2}{A_1 + A_2}$$

$$= 1 - \frac{A_1}{A_1 + A_2}$$

$$= 1 - \frac{(2)(4)}{6 \times 8}$$

$$1 - \frac{1}{6} \quad = \frac{5}{6}$$

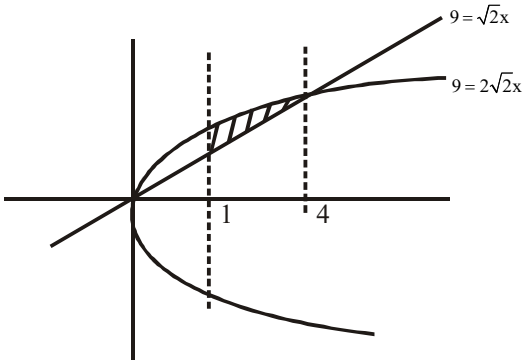
16. The area of the region $S = \left\{ (x, y) : y^2 \le 8x, y \ge \sqrt{2}x, x \ge 1 \right\}$ is : **[JEE(Main)June2022]**

क्षेत्र $S = \left\{ (x, y) : y^2 \le 8x, y \ge \sqrt{2}x, x \ge 1 \right\}$ का क्षेत्रफल है : **[JEE(Main)June2022]**

(A) $\frac{13\sqrt{2}}{6}$ (B) $\frac{11\sqrt{2}}{6}$ (C) $\frac{5\sqrt{2}}{6}$ (D) $\frac{19\sqrt{2}}{6}$

Ans. B

Sol. Required area



$$= \int_1^4 \left(2\sqrt{2x} - \sqrt{2}x\right)dx = \sqrt{2}\left[\frac{4.x^{3/2}}{3} - \frac{x^2}{2}\right]_1^4$$

$$= = \sqrt{2}\left[\frac{32}{3} - 8 - \frac{4}{3} + \frac{1}{2}\right]$$

$$= = \sqrt{2}\left[\frac{64 - 48 - 8 + 3}{6}\right] = \frac{11\sqrt{2}}{6}$$

17. The area of the bounded region enclosed by the curve $y = 3 - \left|x - \frac{1}{2}\right| - |x + 1|$ and the x-axis is :

[JEE(Main)June2022]

वक्र $y = 3 - \left|x - \frac{1}{2}\right| - |x + 1|$ तथा x-अक्ष से घिरे क्षेत्र का क्षेत्रफल है : **[JEE(Main)June2022]**

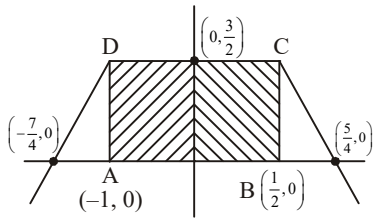
(A) $\frac{4}{9}$ (B) $\frac{45}{16}$ (C) $\frac{27}{8}$ (D) $\frac{63}{16}$

Ans. C

Sol. $y = 3 - \left|x - \frac{1}{2}\right| - |x + 1|$

$$y = \begin{cases} 2x - \frac{7}{2} & x < -1 \\ \frac{3}{2} & -1 \le x \le \frac{1}{2} \\ \frac{5}{2} - 2x & x > \frac{1}{2} \end{cases}$$

graph of y.
required area is area of trapezium ABCD



$= \frac{1}{2}$ [sum of parallel sides] × distance between parallel sides

$= \frac{1}{2}\left(3+\frac{3}{2}\right).\frac{3}{2} = \frac{27}{8}$ sq. units

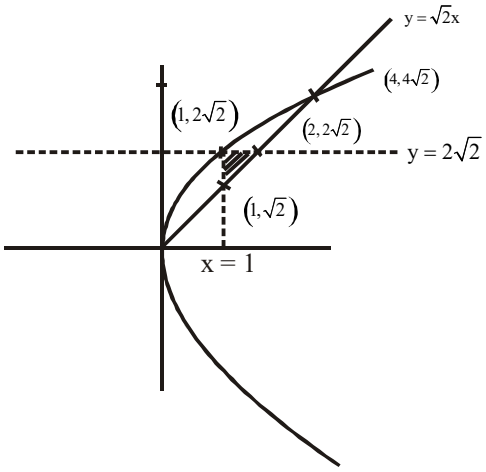
18. The area enclosed by $y^2 = 8x$ and $y = \sqrt{2}x$ that lies outside the triangle formed by $y = \sqrt{2}x$, $x = 1$, $y = 2\sqrt{2}$, is equal to : **[JEE(Main)June2022]**

वक्रों $y^2 = 8x$ तथा $y = \sqrt{2}x$ से घिरे तथा रेखाओं $y = \sqrt{2}x$, $x = 1$, $y = 2\sqrt{2}$ से बने त्रिभुज के बाहर के क्षेत्र का क्षेत्रफल है : **[JEE(Main)June2022]**

(A) $\frac{16\sqrt{2}}{6}$ (B) $\frac{11\sqrt{2}}{6}$ (C) $\frac{13\sqrt{2}}{6}$ (D) $\frac{5\sqrt{2}}{6}$

Ans. C

Sol.



Area of triangle $= \frac{1}{2} \times \sqrt{2} \times 1 = \frac{1}{\sqrt{2}}$

req. area $= \int_0^4 \left(2\sqrt{2}\sqrt{x} - \sqrt{2}x\right) dx - \frac{1}{\sqrt{2}}$

$$= \left(2\sqrt{2}\frac{x^{3/2}}{\frac{3}{2}} - \sqrt{2}\frac{x^2}{2} \right)_0^4 - \frac{1}{\sqrt{2}}$$

$$= \frac{4\sqrt{2}(4)^{3/2}}{3} - \frac{16\sqrt{2}}{2} - \frac{1}{\sqrt{2}}$$

$$= \frac{32\sqrt{2}}{3} - \frac{16\sqrt{2}+\sqrt{2}}{2}$$

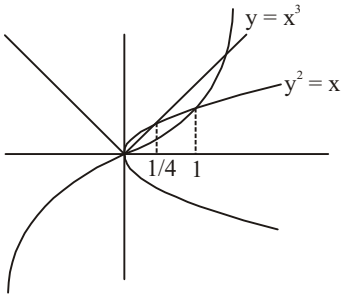
$$= \frac{64\sqrt{2}-51\sqrt{2}}{6} = \frac{13\sqrt{2}}{6} \text{ Ans.}$$

19. Let S be the region bounded by the curves $y = x^3$ and $y^2 = x$. The curve $y = 2|x|$ divides S into two regions of areas R_1 and R_2. If max $\{R_1, R_2\} = R_2$, then $\frac{R_2}{R_1}$ is equal to _____. **[JEE(Main)June2022]**

माना वक्रों $y = x^3$ तथा $y^2 = x$ द्वारा घिरा क्षेत्र S है। वक्र $y = 2|x|$, क्षेत्र S को R_1 तथा R_2 क्षेत्रफल के दो क्षेत्रों में बाँटता है। यदि max $\{R_1, R_2\} = R_2$ है, तो $\frac{R_2}{R_1}$ बराबर है _____ **[JEE(Main)June2022]**

Ans. 19

Sol.



$C_1 : y = x^3$
$C_2 : y^2 = x$ and
C_1 and C_2 intersect at (1, 1)

C_2 and C_3 intersect at $\left(\frac{1}{4}, \frac{1}{2}\right)$

Clearly $R_1 = \int_0^{1/4} \left(\sqrt{x} - 2x\right) dx = \frac{2}{3}\left(\frac{1}{8}\right) - \frac{1}{16} = \frac{1}{48}$ and $R_1 + R_2 = \int_0^1 \left(\sqrt{x} - x^3\right) dx = \frac{2}{3} - \frac{1}{4} = \frac{5}{12}$

so, $\frac{R_1 + R_2}{R_1} = \frac{5/12}{1/48} \Rightarrow 1 + \frac{R_2}{R_1} = 20$

$$\Rightarrow \frac{R_2}{R_1} = 19$$

20. The area (in sq. units) of the region enclosed between the parabola $y^2 = 2x$ and the line $x + y = 4$ is _____.

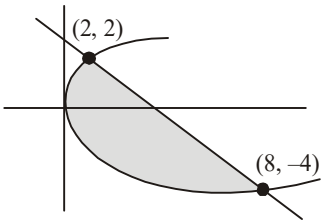
[JEE(Main)June2022]

परवलय $y^2 = 2x$ तथा रेखा $x + y = 4$ से घिरे क्षेत्र का क्षेत्रफल (वर्ग इकाईयों में) है _____

[JEE(Main)June2022]

Ans. 18

Sol.



$$\text{Required area} = \int_{-4}^{2} (x \text{ line} - x \text{ parabola})\, dy$$

$$= \int_{-4}^{2} \left(4 - y - \frac{y^2}{2}\right) dy$$

$$= 4y - \frac{y^2}{2} - \frac{y^3}{6}\Bigg]_{-4}^{2}$$

$$= 18 \text{ square units}$$

21. Let $f(x) = \max\{|x+1|, |x+2|, ..., |x+5|\}$. Then $\int_{-6}^{0} f(x)dx$ is equal to _____.

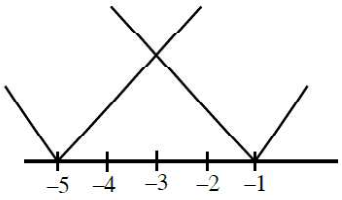
[JEE(Main)June2022]

माना $f(x) = \max\{|x+1|, |x+2|, ..., |x+5|\}$ है। तो $\int_{-6}^{0} f(x)dx$ बराबर है _____

[JEE(Main)June2022]

Ans. 21

Sol. $f(x) = \max\{|x+1|, |x+2|, |x+3|, |x+4|, |x+5|\}$



$$\int_{-6}^{0} f(x)dx = \int_{-6}^{-3} |x+1|dx + \int_{-3}^{0} |x+5|dx$$

$$= -\int_{-6}^{-3} (x+1)dx + \int_{-3}^{0} (x+5)dx$$

$$= -\left[\frac{x^2}{2}+x\right]_{-6}^{-3} + \left[\frac{x^2}{2}+5x\right]_{-3}^{0}$$

$$= -\left[\left(\frac{9}{2}-3\right)-(18-6)\right] + \left[0-\left(\frac{9}{2}-15\right)\right]$$

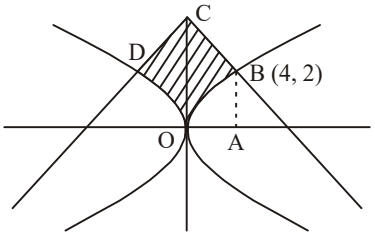
$$= -\left[\frac{3}{2}-12\right] + \frac{21}{2} = \frac{21}{2}+\frac{21}{2} = 21$$

22. Let $A_1 = \{(x,y) : |x| \le y^2, |x|+2y \le 8\}$ and $A_2 = \{(x,y) : |x|+|y| \le k\}$. If 27 (Area A_1) = 5 (Area A_2), then k is equal to _____. **[JEE(Main)June2022]**

माना $A_1 = \{(x,y) : |x| \le y^2, |x|+2y \le 8\}$ तथा $A_2 = \{(x,y) : |x|+|y| \le k\}$ हैं। यदि 27 (क्षेत्रफल A_1) = 5 (क्षेत्रफल A_2) है, तो k बराबर है _____ **[JEE(Main)June2022]**

Ans. 6

Sol. $A_1 = \{(x, y) : |x| \le y^2, |x| + 2y \le 8\}$



for point B $y^2 = x$ & $x + 2y = 8$

$\left(\frac{8-x}{2}\right)^2 = x$ $y = \frac{8-x}{2}$

$x^2 - 16x + 64 = 4x$

$x^2 - 20x + 64 = 0$

$x = 4, y = 2$

$$A_1 = 2\int_0^4 \left(\left(\frac{8-x}{2}\right) - \sqrt{x}\right)dx$$

$$A_1 = 2\left[\frac{8x}{2} - \frac{X^2}{4} - \frac{2x^{\frac{3}{2}}}{3}\right]_0^4$$

$$A_1 = \frac{40}{3}$$

$A_2 = \{(x, y) : |x| + |y| \le k\}$

$A_2 = 4(\text{Area of } \Delta OAB)$

$$= 4 \times \frac{1}{2} k \times k$$

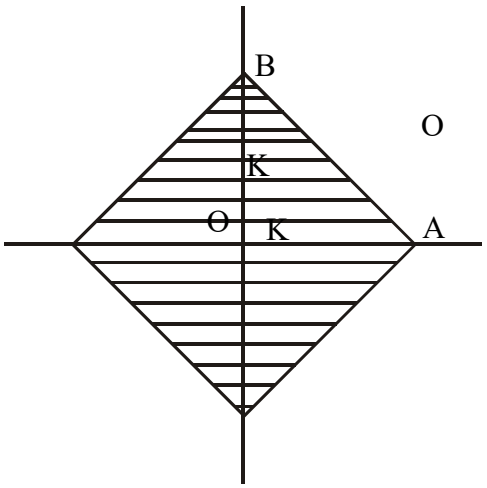
$= 2k^2$

$27(A_1) = 5(A_2)$

$$27 \times \frac{40}{3} = 5 \times 2k^2$$

$k^2 = 36$

$k = 6$



23. If the area of region $\left\{(x, y) : x^{\frac{2}{3}} + y^{\frac{2}{3}} \le 1, x + y \ge 0, y \ge 0\right\}$ is A, then $\frac{256A}{\pi}$ is equal to _____.

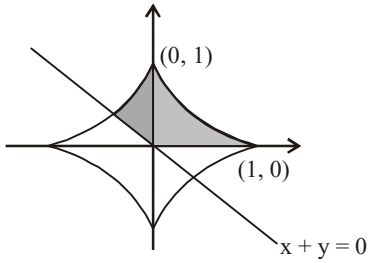
[JEE(Main)June2022]

यदि क्षेत्र $\left\{(x, y) : x^{\frac{2}{3}} + y^{\frac{2}{3}} \le 1, x + y \ge 0, y \ge 0\right\}$ का क्षेत्रफल A है, तो $\frac{256A}{\pi}$ बराबर है _____

[JEE(Main)June2022]

Ans. 36

Sol.



$$A = \frac{3}{2}\int_0^1 \left(1 - x^{\frac{2}{3}}\right)^{\frac{3}{2}} dx$$

put $x = \sin^3\theta$

$$A = \frac{3}{2}\int_0^{\frac{\pi}{2}} \left(1 - \sin^2\theta\right)^{\frac{3}{2}} .3\sin^2\theta\cos\theta\, d\theta$$

$$A = \frac{3}{2}\int_0^{\frac{\pi}{2}} 3\sin^2\theta\cos^4\theta\, d\theta$$

$$A = \frac{9}{2}\times\frac{(1.3)(1)}{(6)(4)(2)}.\frac{\pi}{2}$$

$$A = \frac{9\pi}{64}$$

$$\Rightarrow \frac{64A}{\pi} = 9$$

$$\Rightarrow \frac{256A}{\pi} = 36$$

24. For real numbers a, b (a > b > 0), let Area $\left\{(x,y): x^2+y^2 \le a^2 \ \& \ \frac{x^2}{a^2}+\frac{y^2}{b^2} \ge 1\right\} = 30\pi$ and

Area $\left\{(x,y): x^2+y^2 \ge b^2 \ \& \ \frac{x^2}{a^2}+\frac{y^2}{b^2} \le 1\right\} = 18\pi$. Then the value of $(a-b)^2$ is equal to _____.

[JEE(Main)June2022]

वास्तविक संख्याओं a, b (a > b > 0) के लिए यदि क्षेत्रफल $\left\{(x,y): x^2+y^2 \le a^2 \ \& \ \frac{x^2}{a^2}+\frac{y^2}{b^2} \ge 1\right\} = 30\pi$ तथा

क्षेत्रफल $\left\{(x,y): x^2+y^2 \ge b^2 \ \& \ \frac{x^2}{a^2}+\frac{y^2}{b^2} \le 1\right\} = 18\pi$ हैं, तो $(a-b)^2$ का मान बराबर है _____

[JEE(Main)June2022]

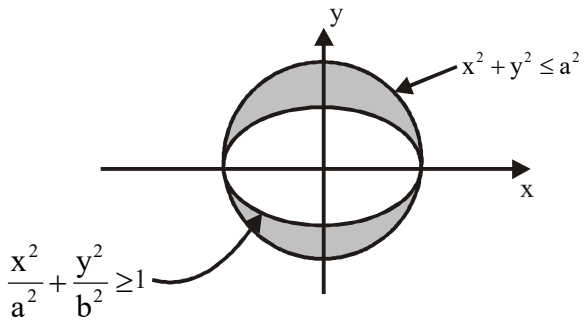
Ans. 12

Sol. Given a,b ∈ R or (a > b > 0)

$$\text{Area}\left\{(x,y): x^2+y^2 \le a^2 \text{and } \frac{x^2}{a^2}+\frac{y^2}{b^2} \le 1\right\} = 30\pi$$

$$\text{Area}\left\{(x,y): x^2+y^2 \le b^2 \text{and } \frac{x^2}{a^2}+\frac{y^2}{b^2} \le 1\right\} = 18\pi$$

Now $x^2+y^2 \le a^2 \ \ \& \ \frac{x^2}{a^2}+\frac{y^2}{b^2} \ge 1$



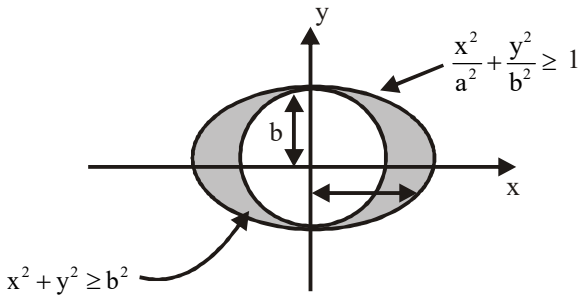
So, shaded region = area of circle - area of ellipse

$\Rightarrow \pi a^2 - \pi ab = 30\pi$

$\Rightarrow \pi a(a-b) = 30\pi$

$\Rightarrow a(a-b) = 30\pi$ .....(1)

Similarly $x^2 + y^2 \ge b^2$ & $\frac{x^2}{a^2} + \frac{y^2}{b^2} \le 1$



So, shaded region = area of ellipse - area of circle

$\Rightarrow \pi ab - \pi b^2 = 18\pi$

$\Rightarrow \pi b(a-b) = 18\pi$

$\Rightarrow b(a-b) = 18\pi$ .....(2)

From (1) – (2)

(a – b) (a – b) = 30 – 18

$(a-b)^2 = 12$

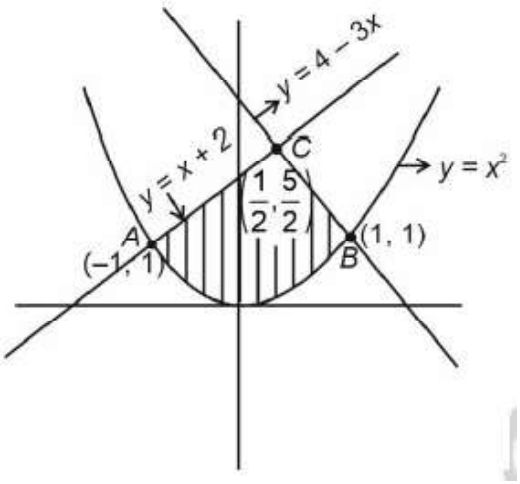
25. The area of the region given by $A = \{(x, y) : x^2 \le y \le \min\{x + 2, 4 - 3x\}$ is : **[JEE(Main)July2022]**

$A = \{(x, y) : x^2 \le y \le \min\{x + 2, 4 - 3x\}$ द्वारा दिए गए क्षेत्र का क्षेत्रफल है : **[JEE(Main)July2022]**

(A) $\frac{31}{8}$ (B) $\frac{17}{6}$ (C) $\frac{19}{6}$ (D) $\frac{27}{8}$

Ans. B

Sol. $A=\left\{(x,y): x^2 \le y \le \min\{x+2, 4-3x\}\right\}$



So area of requierd region

$$A=\int_{-1}^{\frac{1}{2}}\left(x+2-x^2\right)dx+\int_{\frac{1}{2}}^{1}\left(4-3x-x^2\right)dx$$

$$=\left[\frac{x^2}{2}+2x-\frac{x^3}{3}\right]_{-1}^{\frac{1}{2}}+\left[4x-\frac{3x^2}{2}-\frac{x^3}{3}\right]_{\frac{1}{2}}^{1}$$

$$=\left(\frac{1}{8}+1-\frac{1}{24}\right)-\frac{1}{2}-2+\frac{1}{3}+\left(1-\frac{3}{2}-\frac{1}{3}\right)-\left(2-\frac{3}{8}-\frac{1}{24}\right)$$

$$=\frac{17}{6}$$

26. The tangents at the points A(1, 3) and B(1, –1) on the parabola $y^2 - 2x - 2y = 1$ meet at the point P. Then the area (in unit^2) of the triangle PAB is : **[JEE(Main)July2022]**

माना परवलय $y^2 - 2x - 2y = 1$ के बिन्दुओं A(1, 3) तथा B(1, –1) पर स्पर्श रेखाएँ बिन्दु P पर मिलती हैं। तो त्रिभुज PAB का क्षेत्रफल (वर्ग इकाई में) है : **[JEE(Main)July2022]**

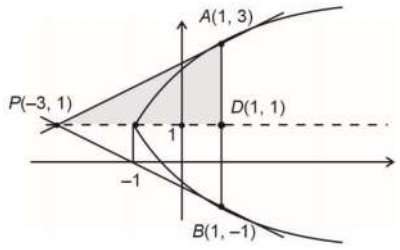
(A) 4 (B) 6 (C) 7 (D) 8

Ans. D

Sol. Given curve : y2 – 2x – 2y = 1.

Can be written as

$(y-1)^2 = 2(x+1)$



And, the given information

Can be plotted as shown in figure

Tangent at A : $2y - x - 5 = 0$ {using $T = 0$}

Intersection with $y = 1$ is $x = -3$

Hence, point P is (–3, 1)

Taking advantage of symmetry

Area of $\Delta PAB = 2 \times \frac{1}{2} \times (1-(-3)) \times (3-1)$

= 8 sq. units

27. The odd natural number a, such that the area of the region bounded by $y = 1$, $y = 3$, $x = 0$, $x = y^a$ is $\frac{364}{3}$, is equal to : **[JEE(Main)July2022]**

विषम पूर्णांक संख्या a, जिसके लिए $y = 1$, $y = 3$, $x = 0$, $x = y^a$ से घिरे क्षेत्र का क्षेत्रफल $\frac{364}{3}$ है, बराबर है:

[JEE(Main)July2022]

(A) 3 (B) 5 (C) 7 (D) 9

Ans. B

Sol. a is a odd natural number and

$$\left|\int_1^3 y^a dy\right| = \frac{364}{3}$$

$$\Rightarrow \left|\frac{1}{a+1}\left(y^{a+1}\right)_1^3\right| = \frac{364}{3}$$

$$\Rightarrow \frac{3^{a+1}-1}{a+1} = \pm\frac{364}{3}$$

Solving with (–) sign,

$\frac{3^{a+1}-1}{a+1} = \frac{-364}{3}$, No a exist

$\therefore$ (a = 5)

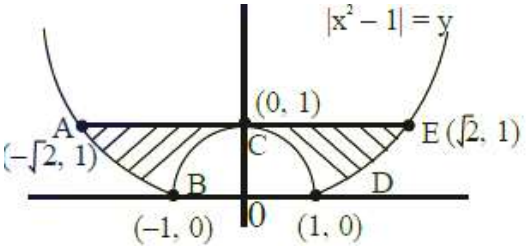
28. The area bounded by the curves $y = |x^2 - 1|$ and $y = 1$ is : **[JEE(Main)July2022]**

वक्रों $y = |x^2 - 1|$ तथा $y = 1$ द्वारा घिरे क्षेत्र का क्षेत्रफल है : **[JEE(Main)July2022]**

(A) $\frac{2}{3}\left(\sqrt{2}+1\right)$ (B) $\frac{4}{3}\left(\sqrt{2}-1\right)$ (C) $2\left(\sqrt{2}-1\right)$ (D) $\frac{8}{3}\left(\sqrt{2}-1\right)$

Ans. D

Sol. $y = |x^2 - 1|$



Arera = ABCDEA

$$= 2\left(\int_0^1 \left(1-\left(1-x^2\right)\right)dx + \int_1^{\sqrt{2}} \left(1-\left(x^2-1\right)\right)dx\right)$$

$$= \frac{8}{3}\left(\sqrt{2}-1\right)$$

29. The area of the smaller region enclosed by the curves $y^2 = 8x + 4$ and $x^2 + y^2 + 4\sqrt{3}x - 4 = 0$ is equal to : **[JEE(Main)July2022]**

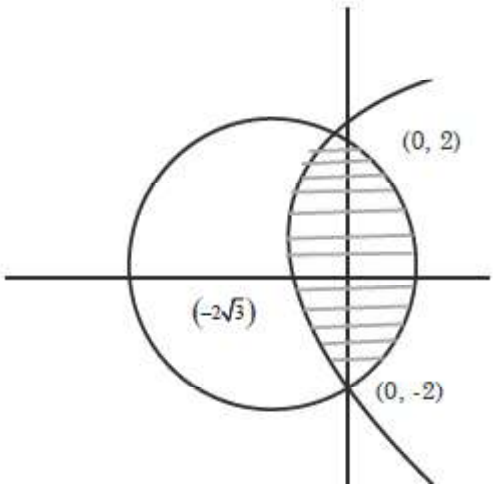
वक्रों $y^2 = 8x + 4$ तथा $x^2 + y^2 + 4\sqrt{3}x - 4 = 0$ से घिरे छोटे क्षेत्र का क्षेत्रफल है : **[JEE(Main)July2022]**

(A) $\frac{1}{3}\left(2-12\sqrt{3}+8\pi\right)$ (B) $\frac{1}{3}\left(2-12\sqrt{3}+6\pi\right)$

(C) $\frac{1}{3}\left(4-12\sqrt{3}+8\pi\right)$ (D) $\frac{1}{3}\left(4-12\sqrt{3}+6\pi\right)$

Ans. C

Sol.



$x^2 + y^2 + 4\sqrt{3}x - 4 = 0$

$y^2 = 8x + 4$

Point of intersections are (0, 2) & (0, –2)

Both are symmetric about x-axis

$$\text{Area} = 2\int_0^2 \left(\sqrt{16 - y^2} - 2\sqrt{3}\right) - \left(\frac{y^2 - 4}{8}\right) dy$$

$$\text{On solving Area} = \frac{1}{3}\left[8\pi + 4 - 12\sqrt{3}\right]$$

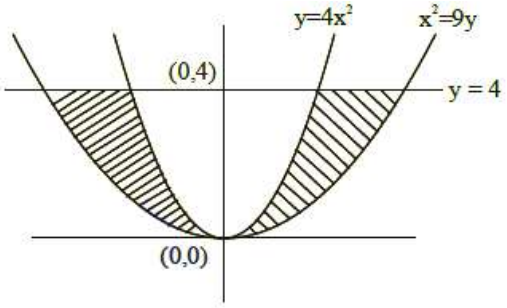
30. The area of the region enclosed by $y \le 4x^2$, $x^2 \le 9y$ and $y \le 4$, is equal to : **[JEE(Main)July2022]**

$y \le 4x^2$, $x^2 \le 9y$ तथा $y \le 4$ घिरे क्षेत्र का क्षेत्रफल है : **[JEE(Main)July2022]**

(A) $\frac{40}{3}$ (B) $\frac{56}{3}$ (C) $\frac{112}{3}$ (D) $\frac{80}{3}$

Ans. D

Sol.



$$\Delta = 2.\int_0^4 \left(3\sqrt{y} - \frac{\sqrt{y}}{2}\right) dy$$

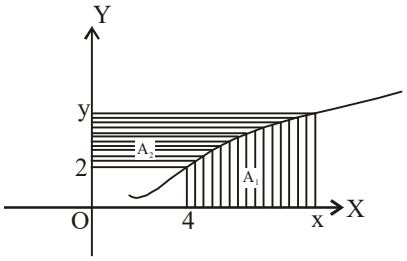
$$= 2.\int_0^4 \frac{5}{2}\sqrt{y}\,dy = \frac{80}{3}$$

31. Consider a curve $y = y(x)$ in the first quadrant as shown in the figure. Let the area A_1 is twice the area A_2. Then the normal to the curve perpendicular to the line $2x - 12y = 15$ does NOT pass through the point :

[JEE(Main)July2022]

चित्र में दिखाए अनुसार $y = y(x)$ प्रथम चतुर्थांश में एक वक्र है। माना क्षेत्रफल A_1, क्षेत्रफल A_2 का दो गुना है। तो वक्र पर, रेखा $2x - 12y = 15$ के लंबवत, अभिलंब किस बिन्दु से होकर नहीं जाता है :

[JEE(Main)July2022]



(A) (6, 21)

(B) (8, 9)

(C) (10, –4)

(D) (12, –15)

Ans. C

Sol. Given that $A_1 = 2A_2$

from the graph $A_1 + A_2 = xy - 8$

$$\Rightarrow \frac{3}{2}A_1 = xy - 8$$

$$\Rightarrow A_1 = \frac{2}{3}xy - \frac{16}{3}$$

$$\Rightarrow \int_4^x f(x)\,dx = \frac{2}{3}xy - \frac{16}{3}$$

$$\Rightarrow f(x) = \frac{2}{3}\left(x\frac{dy}{dx} + y\right)$$

$$\Rightarrow 2\int \frac{dy}{y} = \int \frac{dx}{x}$$

$$\Rightarrow 2\ell ny = \ell nx + \ell nc$$

$$\Rightarrow y^2 = cx$$

As $f(4) = 2 \Rightarrow c = 1$

so $y^2 = x$

slope of normal $= -6$

$$y = -(x) - \frac{1}{2}(-6) - \frac{1}{4}(-6)^3$$

$\Rightarrow y = -6x + 3 + 54$

$\Rightarrow y + 6x = 57$

Now check options and (3) will not satisfy.

32. The area enclosed by the curve $y = \log_e(x + e^2)$, $x = \log_e\left(\frac{2}{y}\right)$ and $x = \log_e 2$, above the line $y = 1$ is :

[JEE(Main)July2022]

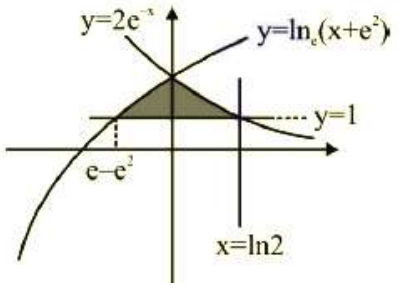
रेखा $y = 1$ के ऊपर वक्रों $y = \log_e(x + e^2)$, $x = \log_e\left(\frac{2}{y}\right)$ तथा $x = \log_e 2$ से घिरे क्षेत्र का क्षेत्रफल है :

[JEE(Main)July2022]

(A) $2 + e - \log_e 2$ (B) $1 + e - \log_e 2$ (C) $e - \log_e 2$ (D) $1 + \log_e 2$

Ans. B

Sol.



Required area is

$$= \int_{e-e^2}^{0} \ell n\left(x + e^2\right) - 1dx + \int_{0}^{\ell n2} 2e^{-x} - 1dx = 1 + e - \ell n2$$

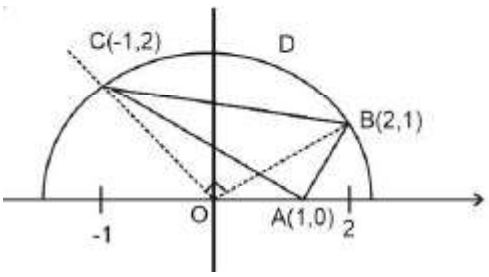
33. The area of the region $\left\{(x, y) : |x - 1| \le y \le \sqrt{5 - x^2}\right\}$ is equal to : **[JEE(Main)July2022]**

क्षेत्र $\left\{(x, y) : |x - 1| \le y \le \sqrt{5 - x^2}\right\}$ का क्षेत्रफल बराबर है : **[JEE(Main)July2022]**

(A) $\frac{5}{2}\sin^{-1}\left(\frac{3}{5}\right) - \frac{1}{2}$ (B) $\frac{5\pi}{4} - \frac{3}{2}$ (C) $\frac{3\pi}{4} + \frac{3}{2}$ (D) $\frac{5\pi}{4} - \frac{1}{2}$

Ans. D

Sol.



$|x-1|< y < \sqrt{5-x^2}$

When $|x-1|=\sqrt{5-x^2}$

$\Rightarrow (x-1)^2 = 5-x^2$

$\Rightarrow x^2 - x - 2 = 0$

$\Rightarrow x = 2, -1$

Required Area = Area of ΔABC + Area of region BCD

$$= \frac{1}{2}\begin{vmatrix} 1 & 0 & 1 \\ 2 & 1 & 1 \\ -1 & 2 & 1 \end{vmatrix} + \frac{\pi}{4}(\sqrt{5})^2 - \frac{1}{2}(\sqrt{5})^2$$

$$= \frac{5\pi}{4} - \frac{1}{2}$$